



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

26-01-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	D	A	A	A	C	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	C	D	B	D	A	A	A	C
SECTION-II	Q.	1	2	3	4	5					
	A.	1.00	180.00	0.25	26.00	30.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	D	B	C	C	B	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	C	B	C	C	D	C	C	A
SECTION-II	Q.	1	2	3	4	5					
	A.	9.00	48.00	3.00	4.00	7.00					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	D	D	C	B	C	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	D	B	C	B	A	C	C	D
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	0.20	0.50	0.50	1.50					



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

26-01-2020

JEE(Main+Advanced) : ENTHUSIAST COURSE (PHASE : I)

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. $y_1 = \sin\left(\omega t + \frac{\pi}{3}\right)$

$$y_2 = \sqrt{2} \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$\text{Phase difference} = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$$

2. Ans. (C)

Sol. $y = 6t - 3.75t^2$

$$v = \frac{dy}{dt} = 6 - 7.5t$$

$$a = \frac{dv}{dt} = -7.5 \text{ m/s}^2$$

$$g_{\text{eff}} = 10 - 7.5 = 2.5 \text{ m/s}^2$$

$$T = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}} \quad \dots(i)$$

$$2 = 2\pi\sqrt{\frac{\ell}{g}}$$

(i)/(ii)

$$T = 4 \text{ sec}$$

3. Ans. (C)

Sol. Centripetal and tangential acceleration both will be there.

4. Ans. (D)

Sol. $t = \frac{a}{v - v \cos \frac{\pi}{n}}$. Here $n = 12$

$$\Rightarrow t = \frac{a}{v - v \cos \frac{2\pi}{12}} = \frac{2a}{v(2 - \sqrt{3})}$$

5. Ans. (A)

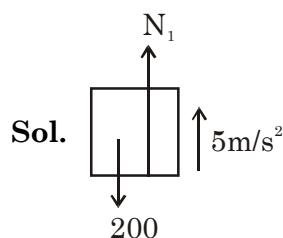
Sol. Time taken by particles to collide

$$t = \sqrt{\frac{2H}{g}}$$

$$\text{Then } u\sqrt{\frac{2H}{g}} + v\sqrt{\frac{2H}{g}} = d$$

$$\Rightarrow u + v = d\sqrt{\frac{g}{2H}} \text{ or } v = d\sqrt{\frac{g}{2H}} - u$$

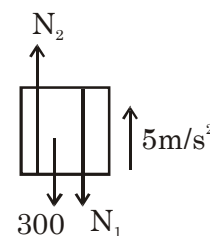
6. Ans. (A)



Sol.

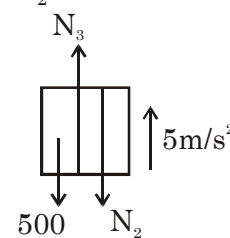
$$N_1 - 200 = 20 \times 5$$

$$N_1 = 300$$



$$N_2 - 300 - 300 = 30 \times 5$$

$$N_2 = 750 \text{ N}$$



$$N_3 - 500 - 750 = 50 \times 5$$

$$N_3 = 1500 \text{ N}$$

7. Ans. (A)

Sol. $T_A = Mg$

$$2T_B = Mg$$

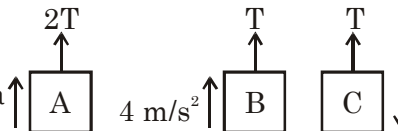
$$\frac{T_A}{2T_B} = 1$$

$$\frac{T_A}{T_B} = 2$$

8. Ans. (C)

Sol. $a = \frac{20-10}{2+1} = \frac{10}{3} \text{ m/s}^2$
 $20 - T = 2a$
 $T = 20 - \frac{20}{3} = \frac{40}{3} \text{ N}$

9. Ans. (B)

Sol. 
 $\Sigma \vec{T} \cdot \vec{a} = 0$
 $2Ta + 4T - 6T = 0$
 $2a = 2$
 $a = 1 \text{ m/s}^2 \text{ (in upward)}$

10. Ans. (C)

Sol. $v^2 = \frac{25}{4} - \frac{1}{4}x^2$
 $v = \frac{1}{2}\sqrt{(5)^2 - x^2}$
 $v = \omega\sqrt{A^2 - x^2}$
 $\omega = \frac{1}{2} \quad A = 5$
 $T = \frac{2\pi}{\omega} = 4\pi$

11. Ans. (B)

Sol. $E_1 = \frac{1}{2}kx^2 \text{ (at } x)$
 $E_2 = \frac{1}{2}ky^2 \text{ (at } y)$
 $E = \frac{1}{2}k(x+y)^2 \text{ (at } x+y)$
 $\sqrt{E} = \sqrt{\frac{k}{2}}\{x+y\}$
 $\sqrt{E} = \sqrt{\frac{k}{2}}\left\{\sqrt{\frac{2E_1}{k}} + \sqrt{\frac{2E_2}{k}}\right\}$
 $\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$

12. Ans. (B)

Sol. $T = 2\pi\sqrt{\frac{k_{eq}}{M}}$
 $k_{eq} = K + 2k + \frac{(2k)(2k)}{2k+2k}$
 $k_{eq} = 4k$
 So $T = 2\pi\sqrt{\frac{4k}{M}}$

13. Ans. (C)

Sol. $m = \frac{15}{15-(-10)} = \frac{15}{25} = \frac{3}{5}$

$\frac{I}{5} = \frac{3}{5}I = 3\text{cm}$

The positive sign indicates that the image is virtual

14. Ans. (D)

Sol. $T = 2\pi\sqrt{\frac{\ell}{g}}$

ℓ = Distance between suspension point and COM.

ℓ is equal in first three cases but maximum for fourth case.

15. Ans. (B)

Sol. $\Delta\phi = (w_1 - w_2)t$
 $= 2\pi\left(\frac{1}{1.5} - \frac{1}{2}\right) \times 1.5 = \frac{\pi}{2}$

16. Ans. (D)

Sol. For disc pendulum

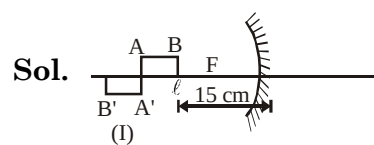
$T = 2\pi\sqrt{\frac{I}{C}} = 2\pi\sqrt{\frac{\frac{3}{2}MR^2}{MgR}}$

$T = 2\pi\sqrt{\frac{3R}{2g}}$

Compare with simple pendulum

$\ell = \frac{3}{2}R$

17. Ans. (A)



For end A : $\rightarrow$

$u_A = -20 \text{ cm}, f = -10 \text{ cm}$

$V_A = \frac{u_A f}{u_A - f} = -20 \text{ cm}$

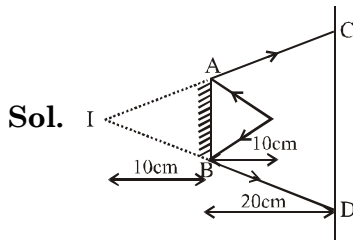
For end B : $\rightarrow$

$u_B = -15 \text{ cm}; f = -10 \text{ cm}$

$V_B = \frac{u_B f}{u_B - f} = \frac{(-15)(-10)}{-15+10} \Rightarrow V_B = -30 \text{ cm}$

Size of image $|\Delta V| = (V_B - V_A) = 10 \text{ cm}$

18. Ans. (A)



$\Delta^S ABI$ & CDI are similar

$$\frac{AB}{CD} = \frac{10}{30} = \frac{1}{3}$$

$$CD = 3(AB) = 3(20) \Rightarrow CD = 60 \text{ cm}$$

$$\text{time} = \frac{\text{Distance}}{\text{Speed}} = \frac{CD}{10} = \frac{60}{10} = 6 \text{ sec.}$$

19. Ans. (A)

Sol. For concave mirror

$$u = +120 \text{ cm}$$

$$f = -60 \text{ cm}$$

$$v = \frac{uf}{v-f} = \frac{(120)(-60)}{120 - (-60)} = -40 \text{ cm}$$

$$m_{II} = -\frac{v}{u} = -\frac{-40}{120} = +\frac{1}{3}$$

For convex mirror :

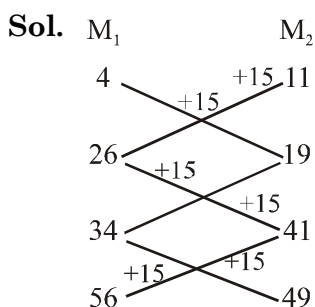
$$u = +(40 - 15) = +25 \text{ cm}$$

$$f = +30 \text{ cm}$$

$$m_2 = \frac{f}{f-u} = \frac{30}{30-25} = +6$$

$$m_{\text{net}} = m_1 \times m_2 = \frac{1}{3} \times 6 = +2$$

20. Ans. (C)



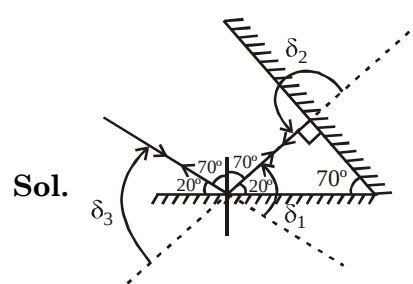
SECTION-II

1. Ans. 1.00

$$\text{Sol. } T = 2\pi \sqrt{\frac{\ell}{g_{\text{eff}}}}$$

$$g_{\text{eff}} = \sqrt{g^2 + a^2}$$

2. Ans. 180.00



Sol.

$$\delta_1 = 180 - 2 \times 70 = 40^\circ (\text{Acos})$$

$$\delta_2 = 180 - 2(0) = 180^\circ (\text{Acos})$$

$$\delta_3 = 180 - 2 \times 70 = 40^\circ (\text{Acos})$$

$$\delta_{\text{net}} = \delta_1 + \delta_2 - \delta_3 = 40 + 180 - 40$$

$$\delta_{\text{net}} = 180^\circ$$

3. Ans. 0.25

Sol. If $u = -\infty$

$$\Rightarrow v = -f \Rightarrow \frac{1}{v^2} = \frac{1}{f^2} = 9$$

$$\Rightarrow a = \frac{1}{f^2} \dots\dots(i)$$

$$\text{if } u = -R \Rightarrow v = -R = -2f \Rightarrow \frac{1}{v^2} = \frac{1}{4f^2}$$

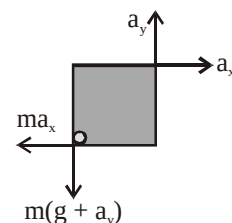
$$\Rightarrow b = \frac{1}{4f^2} \dots\dots(ii)$$

$$\frac{b}{a} = \frac{\frac{1}{4f^2}}{\frac{1}{f^2}} = \frac{1}{4}$$

4. Ans. 26.00

Sol. As $\vec{v} = 5t\hat{i} + 2t\hat{j} \therefore \vec{a} = a_x\hat{i} + a_y\hat{j} = 5\hat{i} + 2\hat{j}$

$$\vec{F} = ma_x\hat{i} + m(g+a_y)\hat{j}$$



$$\therefore |\vec{F}| = m\sqrt{a_x^2 + (g+a_y)^2} = 26 \text{ N}$$

5. Ans. 30.00

Sol. If required angle is θ

From snell's law

$$1 \times \sin 45^\circ = \sqrt{2} \sin \theta$$

$$\sin \theta = \frac{1}{2} \quad \theta = 30^\circ$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)
2. Ans.(B)
3. Ans.(B)
4. Ans.(D)
5. Ans.(B)
6. Ans.(C)
7. Ans.(C)
8. Ans.(B)
9. Ans.(D)
10. Ans.(D)
11. Ans.(C)
12. Ans.(D)

13. Ans.(C)

14. Ans.(B)

15. Ans. (C)

16. Ans. (C)

17. Ans. (D)

18. Ans. (C)

19. Ans. (C)

20. Ans. (A)

SECTION-II

1. Ans.(9.00)

2. Ans.(48.00)

3. Ans.(3.00)

4. Ans.(4.00)

5. Ans.(7.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

$$\text{Sol. } \frac{((x+8)-(8x+1))(\sqrt{5-x}+\sqrt{7x-3})}{((5-x)-(7x-3))(\sqrt{8+x}+\sqrt{8x+1})} = \frac{7}{12}$$

2. Ans. (C)

$$\text{Sol. } f(x) = \lim_{x \rightarrow \infty} \frac{(1+x)}{(2+x)} = 1$$

$$g(x) = \lim_{x \rightarrow \infty} \frac{1}{(1+x)} = 0$$

$$(1)^0 = 1$$

3. Ans. (A)

$$\text{Sol. } \lim_{x \rightarrow 0} 0 \text{ (finite)} = 0$$

4. Ans. (D)

$$\text{Sol. } \lim_{x \rightarrow 2^+} f(g(x)) = f(5+h) = 0$$

$$\lim_{x \rightarrow 2^-} f(g(x)) = f(2-h) = 2 + (2-h)^2 = 6$$

5. Ans. (D)

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{\frac{\ln(1+3x^2)}{3x^2} \times \frac{\sin x}{x} \times 3}{\frac{(3^x-1)}{x} \frac{(1-\cos 2x)}{(2x)^2} \times 4} = \frac{3}{4 \ln 3 \times \frac{1}{2}}$$

6. Ans. (C)

$$\text{Sol. } x = \frac{1}{t} \text{ and solve}$$

7. Ans. (B)

$$\frac{a_{n+1}}{a_n} = \frac{n+4}{n+1}$$

$$\frac{a_2}{a_1} = \frac{5}{2}, \frac{a_3}{a_2} = \frac{6}{3}, \frac{a_4}{a_3} = \frac{7}{4} \dots\dots$$

$$\therefore \frac{a_2}{a_1} \times \frac{a_3}{a_2} \times \frac{a_4}{a_3} \times \dots \times \frac{a_{n+1}}{a_n}$$

$$= \frac{5}{2} \times \frac{6}{3} \times \frac{7}{4} \times \frac{8}{5} \times \frac{9}{6} \times \frac{10}{7} \times \dots \times \frac{n+3}{n} \times \frac{n+4}{n+1}$$

$$a_n = \frac{(n+1)(n+2)(n+3)}{2}$$

$$\therefore \frac{1}{a_n} = \frac{2}{(n+1)(n+2)(n+3)}$$

$$= \frac{1}{n+1} - \frac{2}{n+2} + \frac{1}{n+3}$$

$$\lim_{r \rightarrow \infty} \sum_{k=1}^r \frac{1}{a_k}$$

$$= \lim_{r \rightarrow \infty} \left[\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{2}{4} + \frac{1}{5} \right) + \dots \right]$$

$$= \lim_{r \rightarrow \infty} \left(\frac{1}{6} - \frac{1}{(r+2)(r+3)} \right)$$

$$= \lim_{r \rightarrow \infty} \left(\frac{r^2 + 5r}{6r^2 + 30r + 36} \right) = \frac{1}{6}$$

$$\text{Final Answer is } \frac{5}{6}$$

8. **Ans. (C)**

Sol. $\lim_{x \rightarrow 0} \frac{(6^x - 1)}{x} - \frac{(3^x - 1)}{x} = \ln 6 - \ln 3 = \ln 2$

9. **Ans. (C)**

Sol. Use $f(x) + f(-x) = 0$ and solve

10. **Ans. (B)**

Sol. $f(g(x)) = \ln \left(\frac{1 + \frac{3x + x^3}{1 + 3x^2}}{1 - \frac{3x + x^3}{1 + 3x^2}} \right)$

$$\ln \left(\frac{1+x}{1-x} \right)^3 = 3 \ln \left(\frac{1+x}{1-x} \right) = 3f(x)$$

11. **Ans. (D)**

Sol. $f(0^+) = -\frac{1}{4}$

$$f(0^-) = \frac{2ax}{x(\sqrt{4+ax} + \sqrt{4-ax})} = \frac{a}{2}$$

$$\frac{a}{2} = -\frac{1}{4} \Rightarrow a = -\frac{1}{2}$$

12. **Ans. (D)**

Sol. $r = (1)^\infty = e^{\frac{1}{x}(\sin x - 1)}$

$$\lim_{x \rightarrow 0} e^{\frac{\sin x - x}{x^2}} = e^0 = 1$$

$$m = \lim_{x \rightarrow 0} e^{\frac{1}{x^2}(\frac{\sin x}{x} - 1)} = \lim_{x \rightarrow 0} \frac{(\sin x - x)}{x^3} = e^{-\frac{1}{6}}$$

13. **Ans. (D)**

Sol. $2 \lim_{x \rightarrow 3} f(x) - \lim_{x \rightarrow 3} g(x) = 4$

$$g(3) = 10 - 4 = 6$$

14. **Ans. (B)**

Sol. $\frac{a}{b} = 1; a + 1 = c - 2; 4c - 2 = 0$

$$\frac{12d}{2} = 12 \Rightarrow d = 2$$

$$c = \frac{1}{2}, a = \frac{-5}{2}, b = \frac{-5}{2}$$

15. **Ans. (C)**

Sol. Do yourself

16. **Ans. (B)**

Sol. $x > \sin x \Rightarrow \frac{x}{\sin x} > 1 \Rightarrow \frac{-2x}{\sin x} < -2$

$$\left[\frac{-2x}{\sin x} \right] = -3$$

$$\sin 2x < 2x \Rightarrow \frac{\sin 2x}{x} < 2 \Rightarrow \frac{5 \sin 2x}{x} < 10$$

$$\left[\frac{5 \sin 2x}{x} \right] = 9$$

$$\left[\frac{\tan x}{x} \right] = 1$$

$$k = 7$$

17. **Ans. (A)**

Sol. $k + 1 = 0 \Rightarrow k = -1$

$$\frac{-(1 - \cos \ell x)}{x^2} = -4$$

$$\ell^2 = 8 \Rightarrow \ell = 2\sqrt{2} \text{ or } \ell = -2\sqrt{2}$$

18. **Ans. (C)**

Sol. $p = \lim_{n \rightarrow \infty} \left(e^{1+3+\dots+\frac{n}{2} \text{ terms}} \cdot a^{2+4+\dots+\frac{n}{2} \text{ terms}} \right)^{\frac{1}{n^2+1}}$

$$= \lim_{n \rightarrow \infty} \left(e^{\frac{n}{4}(1+n-1)} \cdot a^{\frac{n}{4}(2+n)} \right)^{\frac{1}{n^2+1}}$$

$$= \lim_{n \rightarrow \infty} \left(e^{\frac{n^2}{4(n^2+1)}} \cdot a^{\frac{(2n+n^2)}{4(n^2+1)}} \right) = e^{\frac{1}{4}} \cdot a^{\frac{1}{4}}$$

$$\Rightarrow p^4 = e \cdot a$$

19. **Ans. (C)**

Sol. $\text{RHL} = \lim_{x \rightarrow 0} \tan^{-1} \left(\frac{1}{x} \right) = \frac{\pi}{2}$

$$\text{LHL} = (-1) \left(-\frac{\pi}{2} \right) = \frac{\pi}{2}$$

20. **Ans. (D)**

Sol. $\lim_{x \rightarrow 0} \frac{2 \sin \frac{3x}{2} \cos \frac{x}{2}}{2 \sin \frac{7x}{2} \cos \frac{x}{2}} = \frac{3}{7}$

SECTION-II

1. **Ans. 2.00**

Sol. Put $x = 1, y = 1$

$$\Rightarrow f(1) = -1 \text{ or } 2 \Rightarrow f(1) = 2$$

$$\text{Put } y = 1$$

$$\Rightarrow f(x) = x + 1 \text{ \& } f^{-1}(x) = (x - 1)$$

$$x^2 - 1 = \tan^{-1} x - 1$$

2. Ans. 0.20

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{(\tan x - \sin x) + \frac{\left(\tan^3 x + \frac{\sin^3 x}{2}\right)}{3} + \dots}{\frac{(\tan x - \sin x)}{x^3} \times x^3} + \dots$$

$$= \lim_{x \rightarrow 0} 1 + \frac{\left(1 + \frac{1}{2}\right)}{3} = 2 = f(0)$$

3. Ans. 0.50

$$\text{Sol. } f\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$f(t) = t^3 - 3t$$

$$f(5) = 125 - 15 = 110$$

4. Ans. 0.50

$$\text{Sol. } \lim_{x \rightarrow 1} \frac{(x+1)}{5(x-2) + \frac{\sin(x-1)}{(x-1)}} = \frac{1}{-5+1} = 0.25$$

$$= \left| \frac{2}{-5+1} \right| = 0.50$$

5. Ans. 1.50

$$\text{Sol. } \lim_{x \rightarrow 1} \frac{(x^2 + ax + b)}{(x-1)(x-2)} = c$$

$$1 + a + b = 0$$

$$\lim_{x \rightarrow 1} \frac{(2x + a)}{(2x - 3)} = c$$

$$a + 2 = c$$

$$a + c + 2 = 0$$

$$2a + b + c = -3$$